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ORIGINAL ARTICLE

Flavonoid rich fraction of *Punica granatum* improves early diabetic nephropathy by ameliorating proteinuria and disturbed glucose homeostasis in experimental animals

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Abstract

Context: Different parts of *Punica granatum* Linn. (Punicaceae) are traditionally used as renal protective agents in the Indian system of medicine. However, there is paucity of information regarding its role in diabetic nephropathy.

Objective: The present study investigates the nephroprotective potential of flavonoid-rich fraction of *P. granatum* leaves in streptozotocin (STZ)-induced early diabetic nephropathy in experimental animals.

Materials and methods: Experimental diabetic nephropathy was induced in Wistar rats by single intraperitoneal injection of STZ (65 mg/kg) dissolved in ice cold citrophosphate buffer (pH 4.3). After induction rats were divided into five groups (6 normal; 24 diabetic) and administered with glibenclamide (5 mg/kg) and three dose levels of flavonoid-rich fraction of *P. granatum* leaves (PGFF), i.e. 50, 100, and 200 mg/kg body weight/day for 28 d. Fasting blood glucose, lipid profile, serum albumin, serum total protein, serum creatinine, blood urea nitrogen (BUN) glycosylated hemoglobin, and biomarkers of kidney oxidative stress were assessed at the end of the treatment period. Urine was analyzed for the measurement of total protein, albumin, and creatinine clearance. Kidney sections were subjected to histopathological study.

Results: Daily oral administration of variable dose levels of PGFF for 28 d normalized various biochemical, metabolic, and histopathological changes in the diabetic rats. PGFF significantly ($p < 0.01$ and $p < 0.05$) improved the glycemic status and renal function in diabetic rats as compared with diabetic control rats.

Discussion and conclusion: The results of our study thus prove the protective effect of PGFF in early diabetic nephropathy by ameliorating proteinuria and disturbed glucose homeostasis in experimental animals.

Keywords

Antihyperglycemic, antihyperlipidemic, antioxidant, quercetin, renal protective, streptozotocin, traditional herbal medicine

History

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Introduction

Diabetic nephropathy as a microvascular disease represents a major long-term complication of diabetes mellitus (Liu et al., 2008; Yang et al., 2013). It is the leading cause of end stage renal disease and accounts for about 30–35% incidents of renal replacement therapy worldwide (Kuhad et al., 2009). Diabetic nephropathy complicates about 30% cases of type I and approximately 15–20% cases of type II diabetes mellitus (Lehmann et al., 2000).

The role of hyperglycemia in the pathogenesis of diabetic nephropathy has been well established in various experimental animal models as well as in human studies (Coimbra et al., 2000). Hyperglycemia is the initiating event which causes structural and functional changes such as glomerular hyperfiltration, glomerular, and tubular epithelial

hypertrophy, and microalbuminuria followed by the development of glomerular basement membrane thickening, accumulation of mesangial matrix, and finally end stage renal disease (Vinod, 2012; Yankunzo et al., 2011). Hyperglycemia is also known to promote oxidative stress and hence involved in the generation of reactive oxygen species (ROS) which play a crucial role in the pathogenesis of diabetic nephropathy (Celik et al., 2009; Luo et al., 2010). Controlling hyperglycemia and improvement of antioxidant capacity will, therefore, prove beneficial in curbing the progression of diabetic nephropathy.

Several plant-derived phytochemicals are known to possess antihyperglycemic and antioxidant activity (Bhutkar & Bhise, 2011). Also, the traditional herbal medicines are widely used to treat diabetes and diabetic complications in Asian countries as the currently available oral hypoglycemic agents have prominent side effects and fail to significantly alter the course of diabetic complications (Balamurugan et al., 2011; Juvekar & Bandawane, 2009). Hence, the traditionally used herbal drugs may prove beneficial in preventing or treating diabetes-induced renal damage.

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Punica granatum Linn. (Punicaceae), commonly known as pomegranate, is an herbal drug-bearing traditional importance for the treatment of diabetes and some renal disorders. In Ayurvedic medicine, pomegranate is considered “a pharmacy unto itself” (Rathod et al., 2012). *Punica granatum* have been prescribed in Unani and Ayurvedic medicines for the treatment of diabetes (Huang et al., 2005).

Punica granatum flower extract is reported to possess antihyperglycemic (Bhaskar et al., 2012; Jafri et al., 2000) and renal protective activity (Singh et al., 2011). Therapeutically beneficial chemical constituents similar to that of flower extract have been reported from *P. granatum* leaves such as ellagitannins, gallic acid, anthocyanins, piperidine alkaloids, flavonoids namely luteolin, apigenin, and quercetin (Garachh et al., 2012; Lansky et al., 2007). Hence, the leaf extract may prove to be beneficial in the treatment of diabetic nephropathy. Further, our earlier work proved the antihyperglycemic and antioxidant activities of *P. granatum* leaves (Patil et al., 2013). Since hyperglycemia and oxidative stress are implicated in the pathogenesis of diabetic nephropathy, we, therefore, hypothesized that *P. granatum* leaves may exert nephroprotective activity through its antihyperglycemic and antioxidant properties. Thus, the aim of our present work was to investigate the nephroprotective effect of flavonoid-rich fraction of *P. granatum* leaves in streptozotocin (STZ)-induced early diabetic nephropathy, to focus on the underlying mechanism of action and to identify the active constituents responsible for the proposed activity.

Materials and methods

Plant material

The leaves of *P. granatum* were collected from the Navsari region of Gujarat state, India, in the month of August 2012. The plant was identified and authenticated by Dr. P. G. Diwakar, Joint Director, Botanical Survey of India, Pune, and a voucher specimen (V. no. USP-1) was deposited in the herbarium for future reference. The leaves were dried in shade and ground to a coarse powder by using a dry grinder.

Animals

Wistar rats of either sex (180–220 g) were procured from National Institute of Bioscience, Chaturshrungi, Pune. All animals were housed in standard polypropylene cages (32.5 × 21 × 14) cm lined with raw husk (renewed after 48 h). The animal house was maintained on 12 h light/dark cycle at approximately 22 ± 2 °C, relative humidity 60–70%, and had free access to standard laboratory diet (Nutrivet Life Sciences, Pune) and water *ad libitum*. The animals were randomly assigned to different groups and a minimum period of 7 d was allowed for the adaptation on each experiment. The animals described as fasting were deprived of food for 24 h before experimentation but allowed free access to water throughout. The study protocol was approved by Institutional Animal Ethical Committee (IAEC) of Modern College of Pharmacy in accordance with the regulations of CPCSEA (884/ac/05/CPCSEA).

Preparation of flavonoid-rich fraction

The leaves powder (100 g) was subjected to cold maceration using solvents methanol:water (70:30) for 72 h to obtain a hydroalcoholic extract. The liquid–liquid partitioning of the crude hydroalcoholic extract was carried out with ethyl acetate (Sannigrahi et al., 2009) to obtain flavonoid-rich fraction (PGFF). The fraction was heated at 40 °C to ensure complete removal of solvent from the fraction. The obtained fraction was stored at 10 °C in the refrigerator until the completion of pharmacological studies.

Chemicals and drugs

STZ was purchased from Sigma-Aldrich (St. Louis, MO). Glibenclamide was procured from Aventis Pharma Ltd (Mumbai, Maharashtra, India). All other chemicals used were of analytical grade.

Quantitative phytochemical analysis

Determination of total phenolic content

The total phenolic content of PGFF was determined according to the method described by Saeed et al. (2012). Briefly, 1 ml of extract/gallic acid (1 mg/ml) was mixed with 1 ml of Folin–Ciocalteu’s phenol reagent. After 5 min, 10 ml of 7% Na₂CO₃ solution was added to the mixture followed by the addition of 13 ml of deionized distilled water and mixed thoroughly. The mixture was kept in the dark for 90 min at 23 °C, after which the absorbance was read at 750 nm. The total phenolic content was determined from the calibration curve of gallic acid solution. The estimation of the phenolic compounds was carried out in triplicate. The total phenolic content was expressed as milligrams of gallic acid equivalents (GAE) per g of dried sample.

Determination of total tannin content

The total tannin content was determined by the hide powder test according to the WHO procedure in which weight difference between tanned and untanned hide powder was used for quantitative determination of tannins (WHO, 1998).

Determination of total flavonoid content

The total flavonoid content was measured by means of an aluminium chloride assay (Zhishen et al., 1999) with slight modification. An aliquot (1 ml) of PGFF (1 mg/ml) or standard solutions of quercetin (10–50 µg/ml) was added to a 10 ml volumetric flask containing 4 ml of 50% solution of methanol. To the flask, 0.3 ml 5% NaNO₂ was added. After 5 min, 0.3 ml of 10% AlCl₃ was added, and at the 6th min, 2 ml NaOH (1 M) solution was added, the total volume was made up to 10 ml with distilled water. The solution was well mixed and the absorbance was measured against the reagent blank at 510 nm. The total flavonoid content (mg/g) was determined from the calibration curve and expressed as mg quercetin equivalents. All the determinations were carried out in triplicate, and the mean values were calculated.

High performance thin layer chromatography (HPTLC)

HPTLC chromatography was performed on 20×10 cm aluminum Lichrosphere HPTLC plates precoated with $200 \mu\text{m}$ layers of silica gel 60F254 (E. Merck, Darmstadt, Germany). PGFF (10 μl) and standard biomarkers quercetin (600 ng/band) and gallic acid (800 ng/band) were applied as bands 6 mm wide and 10 mm apart by means of a Camag Linomat V sample applicator (Muttenez, Switzerland) equipped with a 100 μl syringe. The constant application rate was 160 nl s^{-1} . Linear ascending development with toluene:ethyl acetate:formic acid (5:4:1, v/v/v) as a mobile phase was performed in a 20×10 cm twin-trough glass chamber (Camag) previously saturated with the mobile phase for 15 min at room temperature ($25 \pm 2^\circ\text{C}$) and relative humidity $60\% \pm 5\%$. The development was done using 20 ml of the mobile phase. HPTLC analysis was performed at 270 nm in a reflectance mode with a Camag TLC scanner III operated by WinCATS software (Version 1.2.0, CAMAG Instruments, Farmville, VA). The slit dimensions were 5×0.45 mm and the scanning speed of 20 mm s^{-1} (Husain et al., 2012).

Acute toxicity study

A acute toxicity study was performed according to OECD guidelines no. 423 (OECD, 2008). Female Wistar rats selected by random sampling technique were employed in this study. The animals were fasted overnight with free access to water. PGFF was administered orally to different groups at increasing dose levels of 50, 100, 300, 2000, and 5000 mg/kg body weight. After dosing, the animals were observed for 2 h and then intermittently for further 4 h for changes in behavioral (alertness, restlessness, irritability, and fearfulness), neurological (spontaneous activity, convulsion, gait, bleeding, touch, and pain response), autonomic (defecation and micturition) profiles, and finally recording mortality up to 24 h till 14 d (Mali et al., 2012).

Experimental induction of diabetic nephropathy

Wistar rats (200–240 g) of either sex were fasted overnight before being injected intraperitoneally with a single dose of freshly prepared solution of STZ (STZ, 65 mg/kg) in ice cold citrophosphate buffer (pH 4.3). STZ was first weighed individually for each animal according to their weight and administered within 5 min to prevent its degradation. Since STZ is capable of producing fatal hypoglycemia as a result of massive insulin release, rats were treated with 5% glucose solution for 24 h. Normal control rats received an equivalent volume of citrophosphate buffer. After 7 d of STZ administration, animals showing blood glucose concentration above 200 mg/dl (Bandawane et al., 2011) along with raised urine protein and urine albumin level as compared with normal animals were selected for the study (Yankunzo et al., 2011).

Experimental groups

A total of 36 rats (6 normal; 30 diabetic) were divided into five different groups consisting of six animals in each group as follows.

Group 1: Normal control rats administered cold citrophosphate buffer (pH 4.3)

Group 2: Diabetic control rats, i.e., rats treated with STZ (65 mg/kg, i.p)

Group 3: Diabetic rats treated with 50 mg/kg of PGFF

Group 4: Diabetic rats treated with 100 mg/kg of PGFF

Group 5: Diabetic rats treated with 200 mg/kg of PGFF

Group 6: Diabetic rats treated with 5 mg/kg glibenclamide PGFF as well as glibenclamide were suspended in CMC (0.1%) and administered via oral route using oral feeding needle once daily for 28 d.

Fasting blood glucose

Fasting blood glucose was determined on 0, 7th, 14th, 21st, and 28th day of the study period using a glucometer (Accu Check, Blutzuckermessgeräten, Germany).

Oral glucose tolerance test (OGTT)

The oral glucose tolerance test was performed in overnight fasted (18 h) diabetic rats at the end of 28th day of study. Glucose (3 g/kg) was fed 30 min after the administration of extracts or standard drug. The blood glucose was determined at 0, 30, 60, 90, and 120 min of glucose administration (Bandawane et al., 2013).

Body weight, food, and water intake

During the 28 d study period body weight, food and water intake of animals were recorded every day.

Glycosylated hemoglobin (HbA1c %)

HbA1c% was determined in EDTA–blood samples obtained at the end of 28th day of the study using a commercial assay kit (Crest Biosystems, Goa, India).

Lipid profile

Serum triglycerides (TG), total cholesterol (TC), and high density lipoprotein (HDL) levels were estimated using standard kits (Autozyme Diagnostics, Boisar, Maharashtra, India). Very low density lipoproteins (VLDL) and low density lipoproteins (LDL) levels were calculated using the Friedewald formula (Friedewald et al., 1972).

$$\text{VLDL} = \text{TG}/5$$

$$\text{LDL} = \text{TC} - (\text{HDL} + \text{VLDL})$$

Enzymatic and non-enzymatic biomarkers of oxidative stress in kidney

Estimation of reduced glutathione (GSH)

Reduced glutathione was estimated according to the method described by Kaur et al. (2006). Tissue homogenate (1.0 ml) was precipitated with 1.0 ml of sulfosalicylic acid (4.0%). The samples were kept at 4°C for 1 h and then subjected to centrifugation at 1200 rpm for 15 min at 4°C . The assay mixture contained 0.5 ml of a filtered aliquot, 2.3 ml of sodium phosphate buffer (0.1 M, pH 7.4), and 0.2 ml dithio-bisnitrobenzoate (DTNB) in a total volume of 3.0 ml. The optical density of yellow color thus developed was read immediately at 412 nm. The unknown GSH concentration was

obtained by extrapolation from the standard curve and expressed as U/mg protein.

Estimation of superoxide dismutase (SOD)

The SOD activity was measured according to the method of Marklund (1985). The assay mixture consisted of 2.95 ml Tris-HCl buffer, 25 μ l of pyrogallol, and 0.05 ml of tissue homogenate in a total volume of 3 ml. The difference between the optical densities obtained at 1.30 and 3.30 min was determined and expressed as U/mg protein.

Estimation of catalase (CAT)

Catalase was measured according to the method described by Sahreen et al. (2011). Sample readings were taken by placing 1 ml of phosphate buffer and 5 μ l of tissue homogenate in the reference cuvette and test cuvette. Hydrogen peroxide (10 μ l) was then added in the test cuvette in the spectrophotometer. Reading was taken at 240 nm, 1 min after placing the cuvettes in the spectrophotometer, and expressed as U/mg protein.

Estimation of MDA lipid peroxidation

MDA lipid peroxidation was estimated according to the method described by Kumar et al. (2010). Briefly, the homogenate was incubated with 15% TCA, 0.38% TBA, and 5 N HCl at 95 °C for 15 min. The mixture was cooled, centrifuged, and the absorbance of the supernatant was measured at 532 nm against appropriate blank and expressed as nmol/mg protein.

Parameters of nephroprotective activity

Serum total protein, urinary total protein, serum albumin, and urinary albumin were estimated using a commercial kit (Autozyme Diagnostics, Boisar, Maharashtra, India). Blood urea nitrogen (BUN), serum creatinine, and urinary creatinine clearance were estimated by using a commercial kit (Crest Biosystems, Goa, India). Urine volume was measured at 0, 7th, 14th, 21st, and 28th day of treatment period using metabolic cages. Kidney weight was taken for all the animals at the end of study and the ratio of kidney weight to body weight was determined.

Histopathology study of kidney

At the end of the study, animals were sacrificed and their kidneys were isolated for histopathological estimation. Isolated kidneys were trimmed into small pieces and preserved in phosphate-buffered 10% formaldehyde solution for 24 h. After embedding these pieces in paraffin wax, thin sections of 5 μ m thickness were cut and stained with hematoxylin–eosin dye. The thin sections of kidney were made into permanent slides and examined under high-resolution microscope with photographic facility and photomicrographs were taken (Luna, 1960).

Statistical analysis of data

All the data are presented as mean $\pm$ SEM of measurements made on six animals in each group. Statistical analysis was

performed with a one-way analysis of variance (ANOVA) followed by Dunnet's multiple test of comparison using Graph Pad Instat (version-3) software. A value of $p < 0.05$ was considered to be statistically significant compared with the respective control.

Results

Percentage yield

The yield of hydroalcoholic extract of *P. granatum* leaves was found to be 20.5% w/w while that of PGFF was 8.7% w/w.

Quantitative phytochemical analysis

The total phenolic content of PGFF was found to be 219 ± 1.3 mg GAE/g. The total tannin content of PGFF was found to be 22% w/w while the total flavonoid content of PGFF was found to be 102.02 mg quercetin equivalent/g of fraction.

High performance thin layer chromatography

The overlay spectras of PGFF with gallic acid and PGFF with quercetin are shown in Figures 1 and 2, respectively. Gallic acid and quercetin were resolved at R_f 0.85 and R_f 0.92, respectively. The spectrum clearly reveals that PGFF contains more quercetin than gallic acid.

Acute toxicity study

Results of the acute toxicity study (data not shown) revealed the non-toxic nature of PGFF. No behavioral (alertness, restlessness, irritability, and fearfulness), neurological (spontaneous activity, convulsion, gait, bleeding, touch, and pain response), or autonomic (defecation and micturition) changes were observed during the 14 d observation period. Further, there was no lethality or any toxic reaction in animals at a single large dose of 5000 mg/kg. No mortality was recorded within the 14 d of observation.

Fasting blood glucose

Treatment with PGFF for 4 weeks exhibited a significant ($p < 0.01$) decrease in the fasting blood glucose in STZ-induced diabetic rats as compared with the diabetic control (Figure 3). In diabetic rats, the blood glucose level was reduced by 61.49, 66.21, and 71.68% at 50, 100, and 200 mg/kg doses of PGFF, respectively. The standard oral hypoglycemic drug glibenclamide (5 mg/kg) showed 69.41% reduction in the blood glucose level as compared with the diabetic control group.

Oral glucose tolerance test (OGTT) in diabetic rats

The effect of PGFF on OGTT is shown in Figure 4. The blood glucose of normal and diabetic rats increased significantly ($p < 0.01$) at 30 min after glucose administration. PGFF at all dose levels significantly ($p < 0.01$) reduced the increase in blood glucose at 90 min in glucose-loaded rats compared with diabetic control rats which showed elevation in the blood glucose throughout the total measurement period (120 min).

Figure 1. HPTLC chromatographic overlay spectra of PGFF with standard biomarker gallic acid at 270 nm.

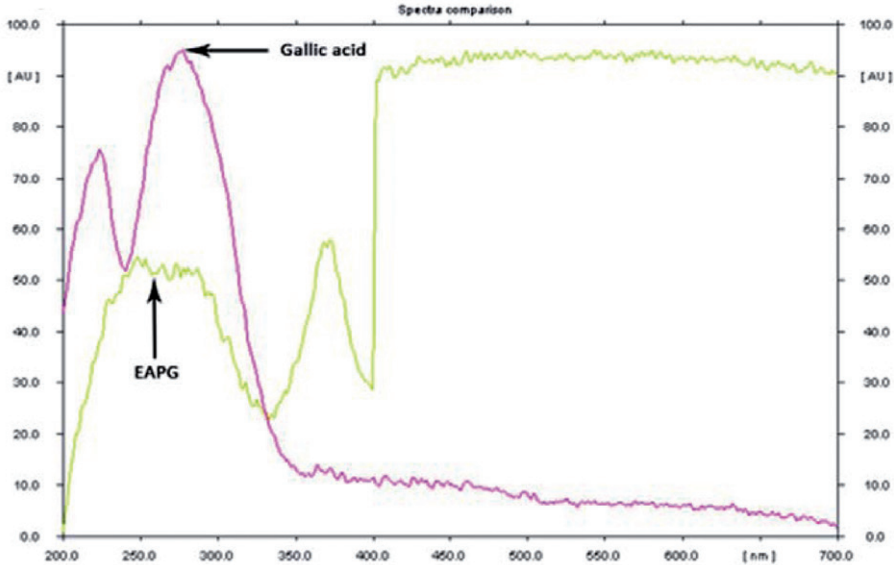


Figure 2. HPTLC chromatographic overlay spectra of PGFF with standard biomarker quercetin at 270 nm.

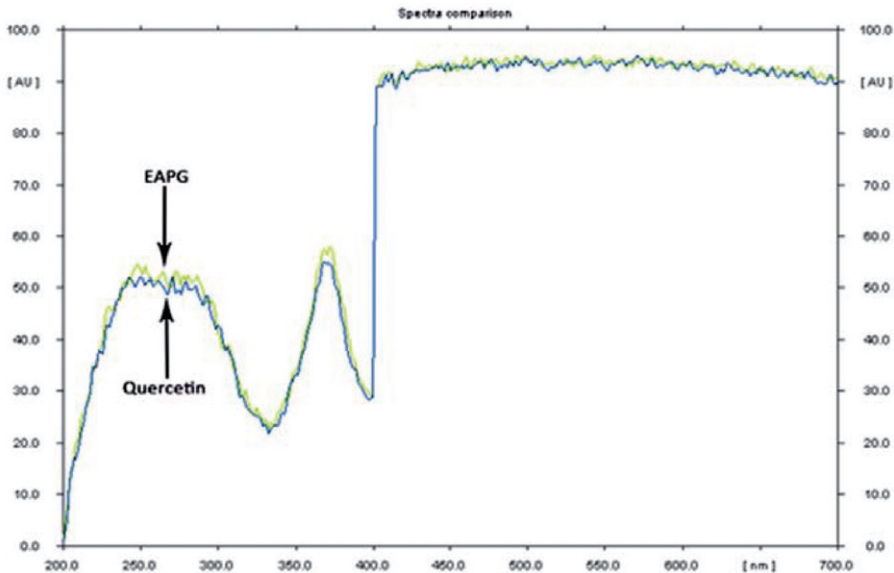


Figure 3. Effect of PGFF on fasting blood glucose in STZ diabetic rats. NC, normal control; DC, diabetic control; PGFF, flavonoid-rich fraction of *Punica granatum* Linn. leaves. Data are expressed as mean $\pm$ S.E.M. ($n = 6$). $\#p < 0.05$, $\#\#p < 0.01$ as compared with NC; $*p < 0.05$, $**p < 0.01$ as compared with DC. Data analyzed by a one-way analysis of variance (ANOVA) followed by Dunnet's multiple test for comparison.

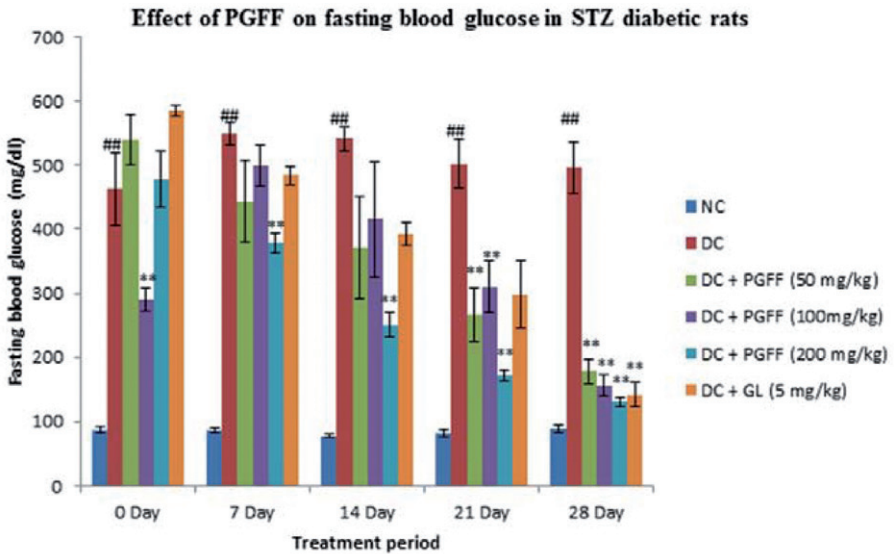


Figure 4. Effect of PGFF on OGTT in STZ diabetic rats. NC, normal control; DC, diabetic control; PGFF, flavonoid-rich fraction of *Punica granatum* Linn. leaves. Data are expressed as mean $\pm$ S.E.M. ($n = 6$). $\#p < 0.05$, $\#\#p < 0.01$ as compared with NC; $*p < 0.05$, $**p < 0.01$ as compared with DC. Data analyzed by a one-way analysis of variance (ANOVA) followed by Dunnett's multiple test for comparison.

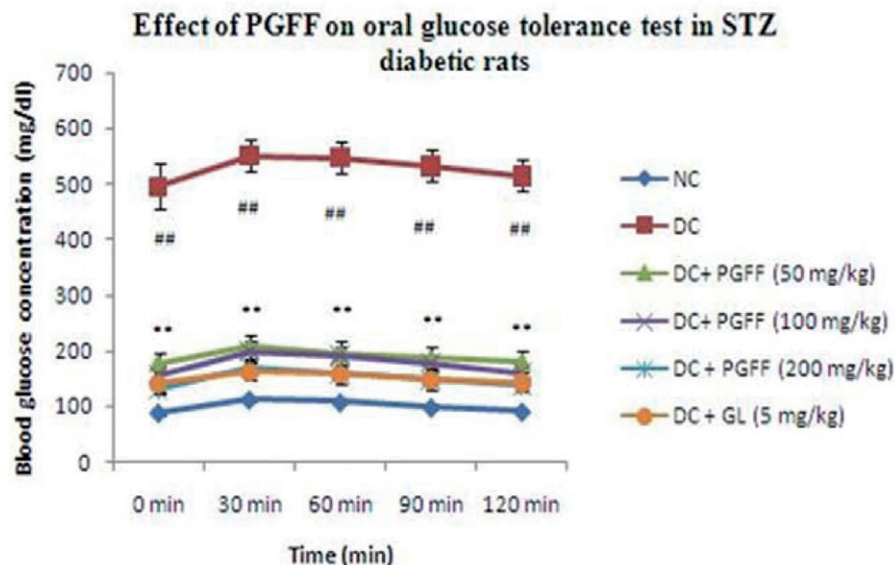
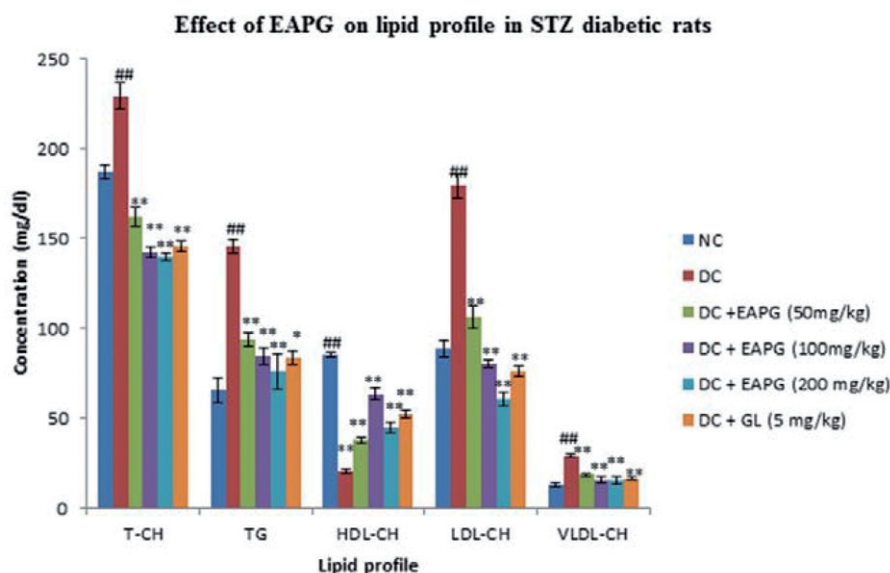


Figure 5. Effect of PGFF on lipid profile in STZ diabetic rats. NC, normal control; DC, diabetic control; PGFF, flavonoid-rich fraction of *Punica granatum* Linn. leaves. Data are expressed as mean $\pm$ S.E.M. ($n = 6$). $\#p < 0.05$, $\#\#p < 0.01$ as compared with NC; $*p < 0.05$, $**p < 0.01$ as compared with DC. Data analyzed by a one-way analysis of variance (ANOVA) followed by Dunnett's multiple test for comparison.



Lipid profile

The protective effect of PGFF on the lipid profile is shown in Figure 5. There was significant ($p < 0.01$) elevation in the levels of T-CH, TG, LDL-CH, and VLDL-CH and significant decrease ($p < 0.01$) in serum HDL-CH when compared with normal rats. PGFF (200 mg/kg) treated diabetic rats decreased T-CH by 39.12%, TG by 47.58%, LDL-CH by 66.34%, and VLDL-CH by 45.99%. Whereas PGFF (200 mg/kg) treated group showed a significant ($p < 0.01$) increase in HDL-CH as compared with the diabetic control group which is comparable with that of the standard drug glibenclamide.

Parameters of nephroprotective activity

As shown in Table 1, glycosylated hemoglobin was significantly ($p < 0.01$) decreased in PGFF-treated diabetic rats as compared with the diabetic control group.

Total protein and albumin were significantly elevated in the urine of the diabetic control group as compared with the normal control group. Treatment with PGFF significantly

($p < 0.01$) reduced urinary total protein and albumin as compared with the diabetic control group. Serum total protein and albumin were also significantly ($p < 0.01$) normalized by PGFF treatment.

BUN and serum creatinine increased significantly ($p < 0.01$) whereas creatinine clearance decreased steeply in diabetic control rats as compared with the normal control group indicating a decreased glomerular filtration rate (GFR). Treatment with PGFF (200 mg/kg) significantly ($p < 0.01$) decreased the alteration in GFR by decreasing serum creatinine, BUN, and increasing creatinine clearance compared with the diabetic control group.

The kidney weight/body weight ratio (KW: BW) of diabetic control rats was significantly higher as compared with other groups which was normalized significantly ($p < 0.01$) by PGFF treatment.

Urine volume of STZ-induced diabetic control rats was found to be significantly ($p < 0.01$) more compared with normal control rats. After 28 d of treatment with PGFF, urine volume significantly ($p < 0.01$) decreased compared with

Table 1. Effect of PGFF on various nephroprotective parameters in STZ diabetic rats.

Parameters	NC	DC	DC + PGFF (50 mg/kg)	DC + PGFF (100 mg/kg)	DC+PGFF (200 mg/kg)	DC + GL (5 mg/kg)
HbA _{1c} (%)	5.41 ± 0.28	7.48 ± 0.04 ^{###}	6.13 ± 0.14 ^{**}	5.91 ± 0.21 ^{**}	6.33 ± 0.19 ^{**}	5.2 ± 0.255 ^{**}
Serum total protein (g/dl)	7.61 ± 0.49	2.99 ± 0.34 ^{###}	5.49 ± 0.429 ^{**}	7.35 ± 0.21 ^{**}	8.253 ± 0.66 ^{**}	8.25 ± 0.66 ^{**}
Serum albumin (g/dl)	9.033 ± 0.321	3.49 ± 0.36 ^{###}	4.718 ± 0.21 ^{**}	5.256 ± 0.129 ^{**}	5.99 ± 0.18 ^{**}	4.233 ± 0.212 ^{**}
Serum creatinine (mg/dl)	6.31 ± 0.69	15.07 ± 0.84 ^{###}	10.98 ± 0.58 ^{**}	10.45 ± 0.57 ^{**}	6.97 ± 0.44 ^{**}	8.91 ± 0.549 [*]
BUN (mg/dl)	20.06 ± 0.98	50.44 ± 3.32 ^{###}	44.30 ± 1.28 [*]	43.34 ± 0.77 [*]	39.62 ± 0.54 ^{**}	39.85 ± 0.94 ^{**}
Urinary total protein (g/dl)	8.14 ± 0.46	12.87 ± 0.44 ^{###}	8.81 ± 0.32 ^{**}	8.48 ± 0.40 ^{**}	7.63 ± 0.42 ^{**}	8.865 ± 0.384 [*]
Urinary albumin (g/L)	0.69 ± 0.03	3.55 ± 0.11 ^{###}	1.73 ± 0.06 ^{**}	1.58 ± 0.06 ^{**}	1.16 ± 0.08 ^{**}	2 ± 0.093 [*]
Urinary creatinine clearance (g/L)	4.96 ± 0.40	0.67 ± 0.87 ^{###}	3.47 ± 0.14 ^{**}	4.20 ± 0.04 ^{**}	4.23 ± 0.03 ^{**}	1.51 ± 0.172
KW:BW	3.02 ± 0.15	7.15 ± 0.62 ^{###}	5.01 ± 0.53 [*]	4.22 ± 0.36 ^{**}	3.40 ± 0.27 ^{**}	4.96 ± 0.87 [*]
Urine volume (ml/24 h)	13.83 ± 0.42 ^{**}	52.16 ± 2.49 ^{###}	16.83 ± 1.35 ^{**}	17.05 ± 0.55 ^{**}	14.83 ± 1.22 ^{**}	20.89 ± 4.86 [*]

NC, normal control; DC, diabetic control; PGFF, flavonoid-rich fraction of *Punica granatum* Linn. leaves. Data are expressed as mean ± S.E.M. ($n = 6$). [#] $p < 0.05$, ^{###} $p < 0.01$ as compared with NC; ^{*} $p < 0.05$, ^{**} $p < 0.01$ as compared with DC. Data analyzed by a one-way analysis of variance (ANOVA) followed by Dunnet's multiple test for comparison.

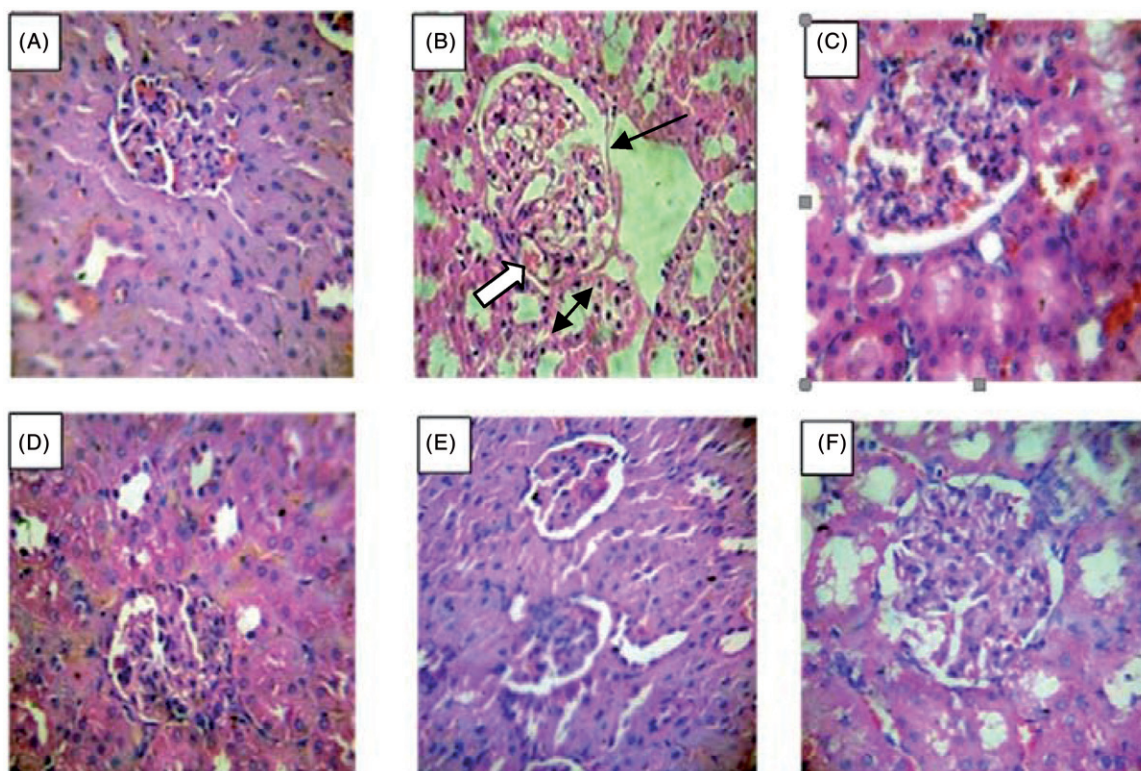


Figure 6. Histopathology of kidney sections of STZ diabetic rats treated with PGFF. Kidney sections were stained with hematoxylin–eosin and observed under 40 × magnification of a digital microscope. (A) Normal control with typical histological structure of rat kidney. (B) Diabetic control group showing glomerular basement membrane thickening (→), tubular dilation (⇔), and cell infiltration (⇔⇔). (C) DC + PGFF (50 mg/kg). (D) DC+PGFF (100 mg/kg). (E) DC + PGFF (200 mg/kg). All PGFF-treated group showed significant improvement in kidney histopathology as compared with DC. (F) Treatment with glibenclamide (5 mg/kg) showed no improvement in the histology of kidney damaged by STZ.

diabetic control groups. However, there was less significant improvement in the renal parameters in diabetic rats treated with glibenclamide.

Enzymatic and non-enzymatic biomarkers of renal oxidative stress

Diabetes resulted in significant decrease in antioxidant enzymes like GSH, SOD, and catalase in kidney homogenate of STZ diabetic rats as shown in Table 2. Moreover, the levels of MDA were significantly increased. PGFF exhibited improvement in antioxidant enzymatic activity compared with the diabetic control group and nearly normalized the

levels of GSH, SOD, MDA, and CAT. Glibenclamide-treated group significantly failed to alter the levels of antioxidant enzymes in diabetic rats.

Body weight, food intake, and water intake

Table 3 shows that the body weight of STZ-induced diabetic control rats was found to be significantly ($p < 0.05$) less compared with normal control rats. After 28 d of treatment with PGFF, the body weight significantly ($p < 0.05$) increased compared with diabetic control groups.

Food intake was significantly high in diabetic control rats as compared with the normal control. At the end of 28 d

Table 2. Effect of PGFF on various renal oxidative stress parameters in STZ diabetic rats.

Parameters	NC	DC	DC + PGFF (50 mg/kg)	DC + PGFF (100 mg/kg)	DC + PGFF (200 mg/kg)	DC + GL (5 mg/kg)
Kidney GSH (U/mg protein)	4.55 ± 1.42	2.73 ± 0.06 ^{##}	3.97 ± 0.06 ^{**}	4.14 ± 0.04 ^{**}	4.53 ± 0.06 ^{**}	3.15 ± 0.005 [*]
Kidney SOD (U/mg protein)	5.671 ± 0.16	3.35 ± 0.16 ^{##}	5.03 ± 0.05 ^{**}	5.71 ± 0.13 ^{**}	6.11 ± 0.12 ^{**}	4.17 ± 0.89 [*]
Kidney catalase (U/mg protein)	78.66 ± 1.68	36.64 ± 1.54 ^{##}	42.59 ± 0.74 ^{**}	46.21 ± 0.65 ^{**}	52.48 ± 0.64 ^{**}	40.093 ± 2.093 [*]
Kidney MDA (nmol/mg protein)	196.92 ± 2.509	389.87 ± 6.58 ^{##}	289.31 ± 4.70 ^{**}	268.55 ± 5.65 ^{**}	229.58 ± 4.03 ^{**}	271.86 ± 9.51 ^{**}

NC, normal control; DC, diabetic control; PGFF, flavonoid-rich fraction of *Punica granatum* Linn. leaves. Data are expressed as mean ± S.E.M. (n = 6). [#]p < 0.05, ^{##}p < 0.01 as compared with NC; ^{*}p < 0.05, ^{**}p < 0.01 as compared with DC. Data analysed by a one-way analysis of variance (ANOVA) followed by Dunnet's multiple test for comparison.

Table 3. Effect of PGFF on body weight, food, and water intake in STZ diabetic rats.

Experimental groups	Body weight (g)		Food intake (g/24 h)		Water intake (ml/24 h)	
	Initial	Final	Initial	Final	Initial	Final
Normal control	189.16 ± 8.78	197 ± 8.45	14.5 ± 0.77	18.08 ± 1.08	14.5 ± 0.77	18.08 ± 1.08
Diabetic control	183.16 ± 7.11	139.16 ± 4.527 [#]	21 ± 1.75 ^{##}	29.75 ± 1.24 ^{##}	41 ± 1.75 ^{##}	59.75 ± 1.24 ^{##}
DC+PGFF (50 mg/kg)	159.66 ± 15.42	178.5 ± 15.84	23.66 ± 0.99	16.33 ± 0.88 ^{**}	48.5 ± 3.60	21.83 ± 1.79 ^{**}
DC+PGFF (100 mg/kg)	176.33 ± 16.22	178.5 ± 15.84 ^{**}	25.16 ± 1.32	13.5 ± 0.99 ^{**}	43.83 ± 1.85	23 ± 1.29 ^{**}
DC+PGFF (200 mg/kg)	191.33 ± 5.66	200.66 ± 3.65 ^{**}	28 ± 3.055 [*]	17.66 ± 1.11 ^{**}	46.16 ± 2.58	18.16 ± 0.79 ^{**}
DC + GL (5 mg/kg)	164.33 ± 5.426	168.16 ± 7.46	18.33 ± 0.909	16.58 ± 0.70 ^{**}	55 ± 2.745 [*]	24.16 ± 3.00 ^{**}

NC, normal control; DC, diabetic control; PGFF, flavonoid-rich fraction of *Punica granatum* Linn. leaves. Data are expressed as mean ± S.E.M. (n = 6). [#]p < 0.05, ^{##}p < 0.01 as compared with NC; ^{*}p < 0.05, ^{**}p < 0.01 as compared with DC. Data analyzed by a one-way analysis of variance (ANOVA) followed by Dunnet's multiple test for comparison.

of treatment food intake of treated groups significantly ($p < 0.01$) decreased as compared with the diabetic control.

A significant decrease ($p < 0.01$) in the water intake was observed in PGFF-treated groups as compared with the diabetic control at the end of the study period.

Histopathology study of kidney

In the normal control group, the glomeruli and tubules appeared normal (Figure 6). In the DC group, matrix expansion was evident in the glomeruli, tubules were dilated and cell infiltrations were observed. Treatment with PGFF inhibited the progression of glomerulosclerosis, tubular dilation, and cell infiltration which was significantly more than glibenclamide treated diabetic rats.

Discussion

In spite of the presence of known antidiabetic medicines in the pharmaceutical market, remedies from medicinal plants are used with success to treat diabetes and its complications. In the present study, protective role of flavonoid-rich fraction of *P. granatum* leaves (PGFF) in early diabetic nephropathy is evaluated through the approaches using STZ-induced diabetic nephropathy in rats. The present results suggest that PGFF exhibit significant antihyperglycemic, hypolipidemic, and nephroprotective effects in STZ diabetic rats.

STZ is widely employed to induce experimental diabetic nephropathy as it selectively destroys the islet of Langerhans (Alhaider et al., 2011). In the present study, induction of diabetic nephropathy by STZ was evidenced by elevated levels of urinary total protein, albumin, serum creatinine, BUN, and decreased creatinine clearance, which were taken

as direct *in vivo* index for nephropathy in STZ diabetic rats (Siddiqui et al., 2010).

Hyperglycemia is considered the main cause of diabetic complications including diabetic nephropathy. Studies have proved that improved glycemic control can significantly decrease the development and progression of diabetic nephropathy (Honore et al., 2012). Antihyperglycemic activity of *P. granatum* leaves is well established (Das & Burman, 2013) and is in agreement with our previous findings (Patil et al., 2013). Therefore, the present study was carried out to study the protective effect of PGFF against early diabetic nephropathy in STZ diabetic rats. In the present study, STZ diabetic rats exhibited significant increase in blood glucose level. Chronic treatment with PGFF-reduced blood glucose level in a duration-dependent manner indicating its potent antihyperglycemic which contributes at least in part in delaying the progression of diabetic nephropathy.

Hyperglycemia leads to an increased production of glomerular matrix proteins, the accumulation of which decreases the surface area for filtration leading to decreased GFR. Decreased GFR is associated with the formation of reactive oxygen species (Craven et al., 1992) and oxidative stress produced by these increased reactive oxygen species generation is a key factor in the development of pathogenesis of diabetic nephropathy (Pan et al., 2009). Hyperglycemia is also known to induce oxidative-nitrosative stress in the rat kidney (Maritim et al., 2003) and this increased oxidative stress in the kidney may induce apoptosis that contribute to the development of diabetic nephropathy (Zhang et al., 2007). Our results are in agreement with these findings and demonstrate that increased production of MDA and decreased GSH, SOD, and CAT activity indicates increased oxidative stress in kidney of STZ diabetic rats. In the present study,

PGFF restored the levels of GSH, SOD, CAT, and MDA close to normal control values which indicate the nephroprotective activity of PGFF through an antioxidant mechanism. Glibenclamide failed to alter the changes in antioxidant defense system which suggests that PGFF may prove to be more beneficial than standard marketed preparation such as glibenclamide in attenuating the progression of diabetic nephropathy.

Renal damage associated with uncontrolled diabetes mellitus show higher concentration of metabolites in blood and urine. This is because kidneys maintain optimum chemical composition of body fluids by removing metabolic wastes such as urea, uric acid, creatinine, and ions (Makni et al., 2010). In the present study, there was an increase in urinary albumin, serum creatinine, and BUN with decreased urinary clearance of creatinine in diabetic rats. Since these are the key biomarkers of diabetes induced renal damage (Lee et al., 2011), decrease in urinary albumin, serum creatinine, and BUN observed in PGFF-treated groups with improvement in urinary clearance of creatinine indicates the ability of PGFF to improve kidney functioning in diabetes mellitus.

Proteinuria, determinant for diabetic renal disease progression, is recognized as a clinical signature, as well as a risk factor of renal lesions in diabetic nephropathy (Parving, 2001). Proteinuria is also associated with a graded increase in the risk of progression to end stage renal disease and cardiovascular event (Lee et al., 1991). A marked increase in urinary protein level was observed in STZ diabetic rats. Significant reduction in urinary protein level after treatment with PGFF in STZ diabetic rats shows that PGFF have the ability to delay the end stage renal disease and associated cardiovascular complications by ameliorating proteinuria.

The effect of PGFF on reducing the risk of cardiovascular and renal complications is further strengthened by studying its effect on lipid profile. Increase in TG, T-CH, VLDL-CH, and LDL-CH in STZ diabetic rats is the result of increased breakdown of lipids, mobilization of free fatty acids from the peripheral stores or increased renal TG synthesis (Ishigaki et al., 2007; Sun et al., 2002). Administration of PGFF for 28 d not only lowered the T-CH, TG, LDL-CH, and VLDL-CH but also enhanced the HDL-CH which is considered to be a protective lipid. Since hyperlipidemia have been linked to renal damage (Wen et al., 2012), we can assume that hypolipidemic action of PGFF may contribute to its nephroprotective activity.

In diabetic condition, exposure of hemoglobin to high levels of plasma glucose causes non-enzymatic glycation of hemoglobin resulting in an increased level of glycosylated hemoglobin. In our study, 7.48% increase in glycosylated hemoglobin was observed in STZ diabetic rats. Reports suggest that elevation of glycosylated hemoglobin beyond 7% generally leads to diabetic nephropathy (Loon et al., 2003). There was marked improvement in glycosylated hemoglobin level by PGFF treatment, which demonstrates its role in delaying the progression of diabetic nephropathy due to its antihyperglycemic activity.

In the present study, decrease in body weight of STZ diabetic rats has been attributed to gluconeogenesis, which is associated with increased muscle wasting and loss of tissue

proteins (Ene et al., 2007; Rajagopal et al., 2008) which is further evident by the reduced levels of serum total protein and albumin in STZ diabetic rats. PGFF significantly ($p < 0.01$) normalized these alterations and may thus act by decreasing gluconeogenesis. A marked increase in kidney weight as observed in our study may be due to hypertrophy caused by early diabetic nephropathy (Tomohiro et al., 2007). Our results in the present investigation showed significant decrease in the kidney weight to body weight ratio in PGFF-treated STZ diabetic rats which confirms renal protective effect of PGFF thereby controlling kidney hypertrophy.

Hyperglycemia is known to increase water and food intake in experimental animals as well as in diabetic patients. In the present study, increase in food and water intake was observed in STZ diabetic rats throughout the study period. Marked reduction in water and feed intake in PGFF-treated rats is because of the glycemic control conferred by PGFF in STZ diabetic rats.

Increase in urine volume is an important indicator of progression to diabetic nephropathy. Urine volume is maintained by the $\text{Na}^+\text{K}^+\text{ATPase}$ pump in the basolateral membrane which hydrolyzes ATP and thus results in transport of Na^+ into the intercellular and interstitial spaces and increases movement of K^+ into cells. The increase in the $\text{Na}^+\text{K}^+\text{ATPase}$ activity in diabetic rat results in increased urine flow (Ghule et al., 2012). We observed an increase in urine volume in STZ diabetic rats. PGFF-treated STZ diabetic rats showed a marked reduction in urine output which may be due to its effect on $\text{Na}^+\text{K}^+\text{ATPase}$.

Renal protective effect of PGFF was further confirmed by the histopathological study of kidney sections. In the present study, kidney sections of untreated STZ diabetic rats showed excessive amassing of the extracellular matrix, thickening of glomerular basement membrane, tubular dilation, and cell infiltration in the kidney tissues. Prevention of the initiation of nephropathy or retardation of the progression of glomerulosclerosis is important in the development of therapeutic agent against renal damage. Treatment with PGFF attenuated extracellular matrix accumulation in the glomerular and tubulointerstitial compartments of kidneys. Reduced glomerular basement membrane thickening, tubular dilation, and cell infiltration into renal tissues by PGFF treatment thus confirms its role as a renal protective agent in STZ-induced diabetic nephropathy.

In our study, the phytoconstituents responsible for the proposed activity were identified by means of quantitative analysis of PGFF via total tannin, total flavonoid, and total phenolic assay which showed the presence of higher flavonoid content than tannin content. HPTLC analysis of PGFF in our study showed that the content of quercetin was comparatively more than gallic acid. Studies by Lansky and Newman (2006) and Wang et al. (2010) have documented the presence of gallic acid and quercetin in *P. granatum* leaves which supports our finding. Further, the protective effects of quercetin in various renal disorders have also been reported (Kanter et al., 2012; Machha et al., 2007). Thus it is clear that the renal protective potential of PGFF in the present study is due to its major flavonoid component quercetin. The gallic acid may act adjuvant with quercetin that probably results in better pharmacological action of PGFF.

Conclusion

In summary, our study showed that PGFF treatment not only alleviated the morphological changes in kidneys of STZ diabetic rats but also improved renal function. The nephro-protective potential of PGFF in STZ diabetic rats is mediated through multiple mechanisms namely antihyperglycemic, hypolipidemic, and antioxidant activity along with amelioration of proteinuria and improvement in structural integrity of renal tissues which may be due to its major component quercetin and gallic acid. The protective effect of PGFF against diabetes induced renal damage was comparably more than that of standard marketed antidiabetic agent such as glibenclamide. However, further studies are required for the clarification of detailed mechanism of renal protective action of *P. granatum* leaves which will provide new therapeutic targets for the treatment of diabetic nephropathy.

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Declaration of interest

The authors report no declarations of interest.

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